

Question

Chromatography is a good method for separating and identifying compounds. In chromatography, substances with different properties can be separated by the different partition ratios of compounds in the stationary phase/mobile phase. A very important value for judging the separation efficiency of chromatography is the R_f value (retardation factor), which is the ratio of the distance of the separated substance to the origin to the distance of the solvent front to the origin. In addition to changing the developing solvent, there is also a common technique in the laboratory to separate two compounds on a thin-layer chromatography plate to clarify the composition of the system: after performing one *TLC* separation, the solvent is evaporated, and then another development is performed (assuming the solvent front reaches the same height). It is known that there are two compounds in the system, and their R_f values in the developing solvent are a and b respectively ($0 < a < b < 1$).

The following statements are correct:

1. To achieve better separation with secondary development, it is required that $a < b < 0.5$
2. If $a=0.3$, $b=0.6$, the separation effect of secondary development is better than that of single development.
3. If $a=0.3$, $b=0.4$, the separation effect of tertiary development is better than that of secondary development.
4. If $a=0.2$, $b=0.3$, the separation effect of tertiary development is better than that of secondary development.
5. Tertiary separation is better than secondary separation, requiring $a + b < 0.67$
6. Tertiary development achieves better separation requires: $a^2 + b^2 + ab - 3a - 3b + 1 > 0$

A. All other options are incorrect

B. 2.4.6.

C. 1.3.4.

D. 2.4.5.

E. 3.4.

F. 4.5.

G. 2.6.

H. 2.4.

Answer

Correct Answer: **H**

Detailed Explanation

Assume the moving distance of the solvent front is L , then the moving distance of the first compound is aL , and the moving distance of the second compound is $(1 - a)L$.

$$Rf' = 2a - a^2$$

Better separation effect refers to being separated further apart in the TLC plate, with a larger difference in Rf values.

$$\Delta Rf' = -2a + 2b - b^2 + a^2 = (b - a)(2 - a - b) > b - a$$

Then the requirement is: $a + b < 1$

CHECKPOINT

1 PTS

The requirement for better separation effect with double development: $a + b < 1$

Statement 1. Using double development to achieve a better separation effect requires $a < b < 0.5$, which does not conform.

In statement 2, if $a=0.3$, $b=0.6$, $a+b<1$, which is correct, the separation effect of double development is better than that of single development.

Statement 1 is incorrect, and statement 2 is correct.

CHECKPOINT

0.5 PTS

Statement 1 is incorrect

CHECKPOINT

0.5 PTS

Statement 2 is correct

For triple development, $Rf' = 1 - (1 - a)^3 = 3a - 3a^2 + a^3$

To make the separation effect of the triple development method better, it needs to satisfy:

$$\Delta Rf' = (b - a)(3 - 3b - 3a + b^2 + a^2 + ab) > (b - a)(2 - a - b)$$

The condition can be obtained as $a^2 + b^2 + ab - 2a - 2b + 1 > 0$, or rearranged as $(a + b - 1)^2 > ab$

CHECKPOINT

1 PTS

The requirement for better separation effect with triple development: $a^2 + b^2 + ab - 2a - 2b + 1 > 0$, or rearranged as $(a + b - 1)^2 > ab$

If $a=0.3$, $b=0.4$, the separation effect of triple development is better than that of double development. $a^2 + b^2 + ab - 2a - 2b + 1 = -0.03 < 0$ Statement 3 is negative when substituted, incorrect.

If $a=0.2$, $b=0.3$, the separation effect of triple development is better than that of double development. $a^2 + b^2 + ab - 2a - 2b + 1 = 0.19 > 0$ Statement 4 is positive when substituted, correct.

CHECKPOINT

0.5 PTS

Statement 3 is incorrect

Statement 4 is positive, correct.

CHECKPOINT

0.5 PTS

Statement 4 is correct

Statement 5 is not equivalent to the conclusion, and in statement 6, the coefficient before a and b changes from 2 to 3, both are incorrect.

CHECKPOINT

0.5 PTS

Statement 5 is incorrect

CHECKPOINT

0.5 PTS

Statement 6 is incorrect

Therefore, statements 2 and 4 are correct, choose H